

Local Docker Run Guide

Clone from GitHub and run the Spring Boot URL shortener with Docker Compose

Purpose

Copy the project from GitHub to a local Windows machine and run PostgreSQL, Redis, and the Spring Boot application with Docker Compose.

Prerequisites

- ? Git installed and available in PowerShell.
- ? Docker Desktop installed and running.
- ? Internet access to pull images and clone the repository.
- ? Optional: Java 21 and Gradle 8.x for non-Docker Gradle commands.

Step 1 - Clone the Project

```
cd C:\Users\de11\Documents\Codex
git clone https://github.com/saurabhbarnwal/ai-assisted-url-shortener.git
cd ai-assisted-url-shortener
```

Step 2 - Confirm Files

File	Purpose
Dockerfile	Builds the Spring Boot app image.
docker-compose.yml	Starts app, PostgreSQL, and Redis together.
build.gradle	Defines Java 21, Spring Boot, and dependencies.
src/main/resources/application.yml	Defines local defaults for PostgreSQL and Redis.

Step 3 - Start the Application

```
docker compose up --build
```

Expected containers:

Service	Purpose	Port
app	Spring Boot URL shortener	8080
postgres	PostgreSQL database	5432
redis	Redis cache	6379

Step 4 - Verify Health

```
Invoke-RestMethod http://localhost:8080/actuator/health
```

Expected response:

```
{"status": "UP"}
```

Step 5 - Test APIs

Create a short URL:

```
Invoke-RestMethod `
-Method Post `
-Uri http://localhost:8080/api/v1/urls `
```

```
-ContentType application/json `
-Body '{"originalUrl":"https://www.google.com"}'
```

Use the shortCode from the response:

```
curl.exe -i http://localhost:8080/YOUR_SHORT_CODE
```

```
Invoke-RestMethod http://localhost:8080/api/v1/analytics/YOUR_SHORT_CODE
```

Step 6 - Swagger UI

```
http://localhost:8080/swagger-ui.html
```

Step 7 - Stop or Reset

```
docker compose down
```

```
# Delete local database volume too
docker compose down -v
```

Optional Tests

```
gradle clean test
```

```
# Docker-backed tests
gradle clean integrationTest --stacktrace --info
```

Troubleshooting

Problem	Check
Port 8080 already in use	Stop the other process or change compose port mapping.
PostgreSQL connection fails	Run docker compose ps and confirm postgres is healthy.
Redis connection fails	Run docker compose ps and confirm redis is healthy.
Testcontainers fails	Confirm Docker Desktop is running and docker info succeeds.
Swagger not loading	Confirm /actuator/health returns UP.

Local Environment Variables

Variable	Default in Compose
SPRING_DATASOURCE_URL	jdbc:postgresql://postgres:5432/urlshortener
SPRING_DATASOURCE_USERNAME	urlshortener
SPRING_DATASOURCE_PASSWORD	urlshortener
SPRING_DATA_REDIS_HOST	redis
SPRING_DATA_REDIS_PORT	6379
APP_BASE_URL	http://localhost:8080